ZigBee Protocol Fuzzing

Mengfei Ren, Ming Jiang, Yu Lei
Department of Computer Science and Engineering, University of Texas at Arlington

Introduction

Recent years the Internet of Things (IoT) devices have rapidly grown and been adopted by various areas. The security detection also become imperative since various previous attacks launched on personal computers have been deployed on IoT devices as well. As Figure 1 shown, these attacks target not only embedded device, but also software and application that remotely control the IoT device as well as network communication between device and application.

Our research focus on security issues of the network protocol especially the low power protocols that have special design and stack layout. In this paper, we present a fuzzing approach to detect vulnerabilities in ZigBee protocol, which one of popular low power protocols in IoT network. Instead focusing on test input generation, we add the power of coverage-guided fuzzing to generative fuzzing approach to improve its mutation effectiveness.

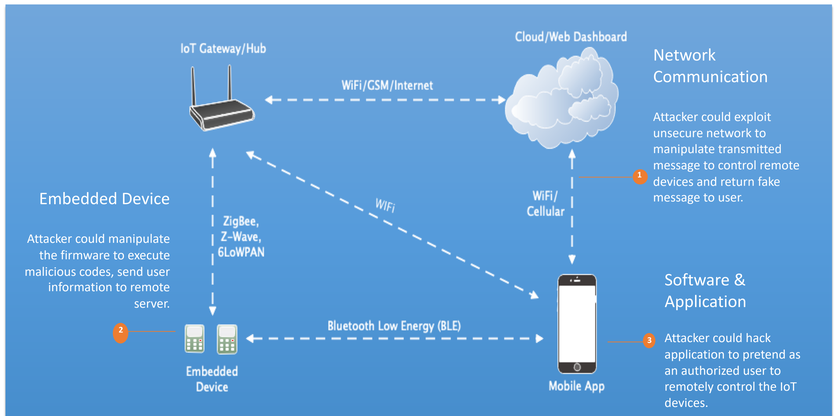


Figure 1. Attacks Surface in IoT Network

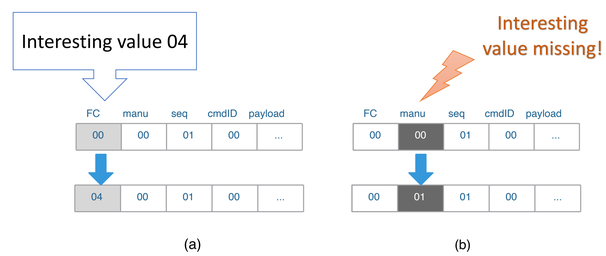
ZigBee Protocol

ZigBee protocol is a wireless personal area network (PAN) standard for resource constrained network. ZigBee Alliance maintains the protocol specification and defines network management, security and application layer. ZigBee protocol stack is build upon IEEE 802.15.4 which defines MAC layer and PHY layer.

IoT device manufacturer vendors build their own device upon the protocol stack. To perform device functionalities, vendors could use their specific library or the general library provide by ZigBee protocol, which is called ZigBee Cluster Library (ZCL). ZCL defines a library of application capabilities over a diverse range of market vendors so devices developed by different vendors are compatible to each other.

Observations

1. IoT device testing usually requires firmware or many physical devices setting up.
2. Protocol fuzzing is generation-based fuzzing that mutates message field individually so that some interesting values are missing during mutation.



Questions

Q1. Could we mutate test input based on interesting value in generative fuzzing approach, like how mutation-based fuzzing does?
Q2. Could we fuzz ZigBee protocol like regular PC software and protocols in OSI model?

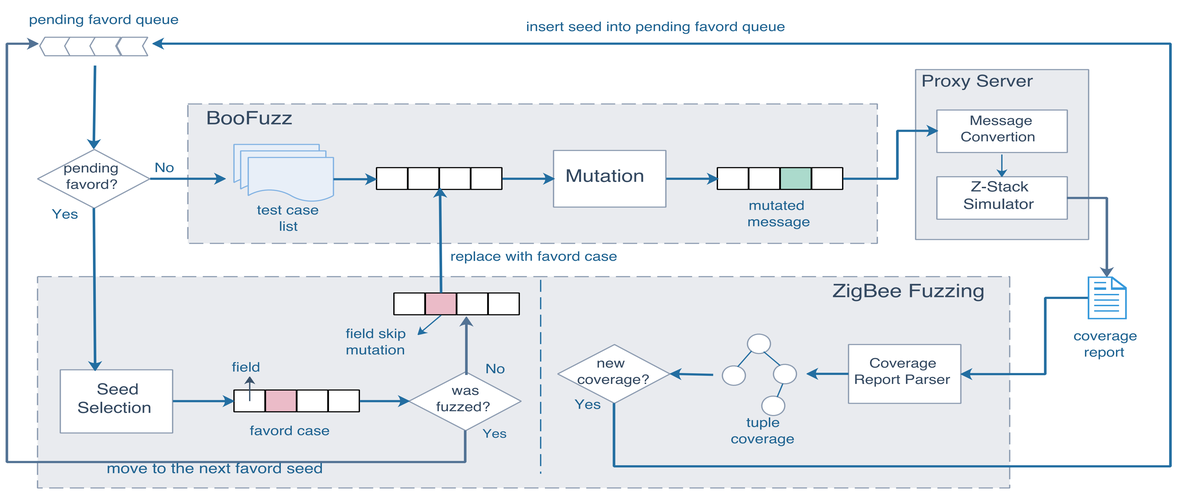
Approach

1. The framework is built on BooFuzz, a state-of-art protocol fuzzing framework which is a successor of Sulley.
2. **No embedded devices and firmware required** – a pure software simulation environment to execute ZigBee protocol including coverage generation.
3. **Coverage-guided generative fuzzing** – We use coverage as heuristic to record interesting values then fuzz test inputs based on those values.

Preliminary Results

	BooFuzz	Our approach
Total Crash Found	153	337
Statement coverage	865 (55.81%)	875 (56.45%)
Tuple coverage	563	585
Total test case generated	6428	16225
Index of test case first finds Bug1*	300	18
Index of test case first finds Bug2*	2782	2517
Index for test case first finds Bug3*	1609	1325

ZigBee Protocol Fuzzing Framework



Contact

Mengfei Ren, email: mengfei.ren@mavs.uta.edu
Jiang Ming, email: jiang.ming@uta.edu
Yu Lei, email: ylei@cse.uta.edu

* This work is still in progress so we cannot release the real bug names right now.

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